

Machine Learning Project Report

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1 Task Overview

In this project, we work on UCI Mushroom dataset. It contains 8124 rows of mushroom records with 22 categorical features, which describe biological characteristics of each mushroom. The objective of the project is to predict whether a mushroom is edible, or poisonous.

The difficulty of the task lies in the categorical nature of the features, which requires complex preprocessing and a fitting encoding strategy. Missing values, indicated by "?", must also be accounted for. Lastly, the linear separability of the dataset is not obvious, which may require a more complex model.

2 Methods

In order to solve the classification task, 3 methods have been studied in detail.

1. Ridge classification: add a penalty on the ℓ^2 -norm of the coefficients:

$$\min_w \frac{1}{2N} \|Xw - Y\|_2^2 + \lambda \|w\|_2^2.$$

2. Linear SVM: find the best separating hyperplane, using the maximum margin principle. That is, for a hyperplane

$$H = \{x \in \mathbb{R}^d \mid \langle w, x \rangle + b = 0\}, \quad w \in \mathbb{R}^d, b \in \mathbb{R},$$

we want to maximize

$$\text{dist}(x, H) = \frac{|\langle w, x \rangle + b|}{\|w\|_2}.$$

3. SVM with an RBF kernel:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right).$$

Here, choosing the parameters σ and C (appears in the formulation of the soft-margin SVM optimization problem), is of main importance, and we will conduct a study to find the pair, yielding the best performance.

3 Experiments and Results

To test the first and second models, 3 encoding strategies have been used:

- one-hot encoding: $\text{OneHot}_k(x) = \mathbb{1}\{x = c_k\}$
- ordinal encoding: $\text{Ordinal}(x = c_k) = \text{rank}(c_k)$
- frequency encoding: $\text{Freq}(x = c_k) = \frac{\text{count}(x=c_k)}{N}$

Initialized at $w = 0$ and $b = 0$, the models trained on 20% of the entire dataset, over 5 folds (for cross-validation). The visualized results can be found below.

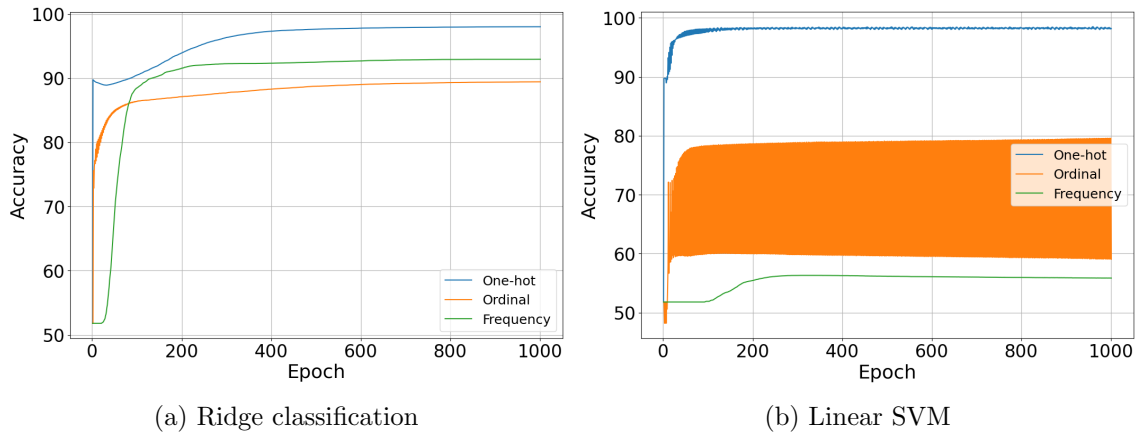


Figure 1: Mushroom edibility prediction accuracies

Additionally, a more complicated model, namely the RBF kernel method, has been implemented in hope of achieving a better performance. The training has been conducted for numerous pairs of $(\sigma; C)$ parameters. The graph, depicted below, represents the common "Accuracy vs. Complexity" trade-off (the amount of support vectors corresponds to the complexity of the model, more vectors $\equiv$ harder to find a decision boundary).

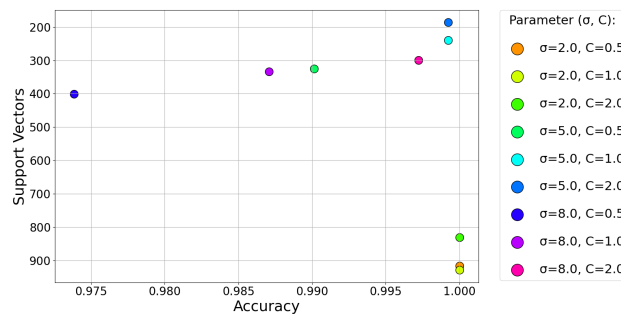


Figure 2: Kernel SVM Accuracy vs. Complexity Trade-Off

4 Discussion

In the plots below it is illustrated, how different encoding methods treat the raw data:

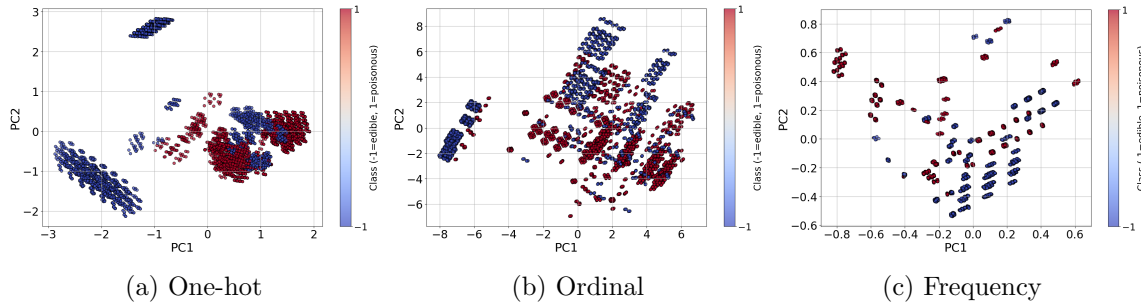


Figure 3: PCA of differently encoded features

- One-hot encoding treats every category on its own, hence one sees two colors separately quite clearly. This is being reflected in both linear SVM and Ridge classification accuracies (Figure 1) being stable and going up to 98%.
- While working on the ordinally-encoded dataset, the linear SVM method gets confused by the unstable decision boundaries and is hence very shaky (ordinal encoding imposes a numeric order on the categorical data, which does not fit here; the binary categories also cannot be clearly separated on the PCA graph, and have a meaningless direction). Ridge classification, due to its smaller sensitivity towards poorly scaled or ordered features, performs better and achieves an accuracy rate of $\approx 88\%$.
- Lastly, frequency-encoded data becomes hard to separate at all, and on the PCA graph, the categories are densely intertwined. Hence, the linear SVM model accuracy becomes stable at $\approx 50\%$ and cannot improve further (the model ends up "guessing" the category). Ridge classification, similarly to the ordinal case, manages to reach an accuracy rate of $\approx 93\%$ (again, due to its simplicity, comparing to the linear SVM model).

In summary, one can see that both Ridge classification and linear SVM models, applied to one-hot encoded data, perform best due to the mushroom dataset's clear decision boundary and linear separability. Other encoding methods do not fit the model - Ridge classification, due to its lower sensitivity, still manages to reach good performance, while a more complex linear SVM model struggles to find a clear decision boundary. To round up the study, a more complex method (RBF kernel SVM) has been applied to the one-hot encoded dataset (since it fits the mushroom dataset best) and compared to the linear SVM and Ridge methods from earlier. Since its performance depends massively on the choice of parameters σ and C , we experimented with different pairs to find the optimal setting. As depicted in Figure 2, a pair of $\sigma = 5.0$ and $C = 1.0$ provided an accuracy rate of 99.5%, while keeping the model relatively simple at ≈ 200 support vectors.

Compared to SVM and Ridge classification on tin sine one-hot encoded data, both of which achieved an accuracy rate of 98%, the RBF kernel slightly improved accuracy, but at the cost of increased complexity and training time. Linear SVM and Ridge classifiers are simpler, faster to train, and require no parameter tuning, making them more practical choices when the data is nearly linearly separable, which is the case for the mushroom dataset.